Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: 20260205_171124_small_image_classifier.pt

Attack Type: FGSM
Images Evaluated: 10

Attack Success Rate	Original Confidence	Adversarial Confidence	Execution Time
100.0%	36.89%	48.71%	3.66s
			Accuracy drop: -11.82%

Threat Level Assessment

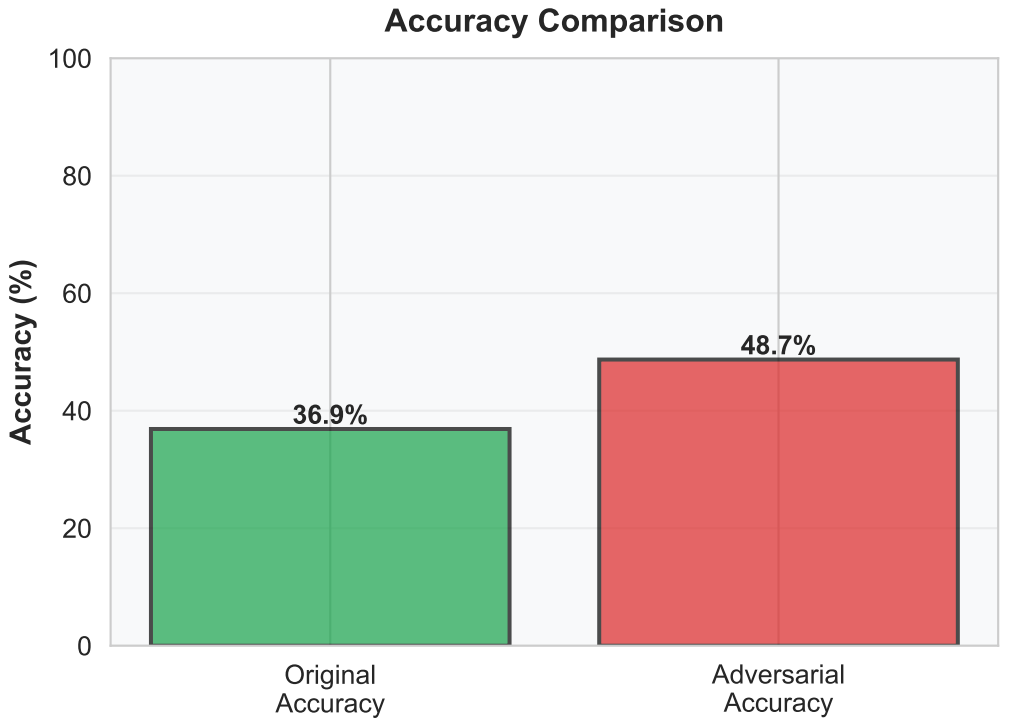
HIGH RISK

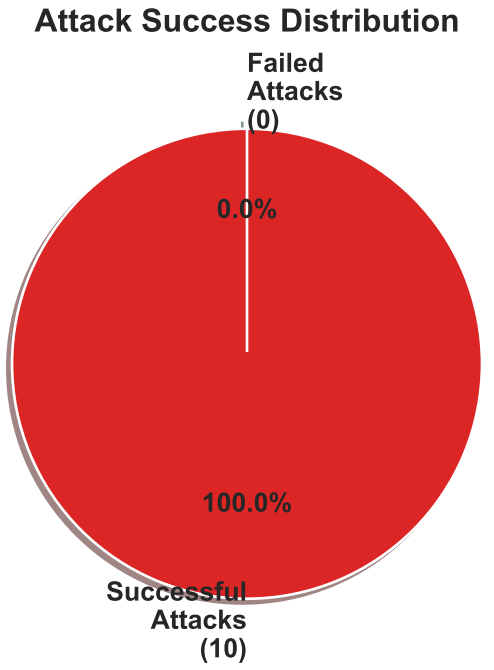
The FGSM attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 36.89% to 48.71%, indicating the current robustness posture is high risk.

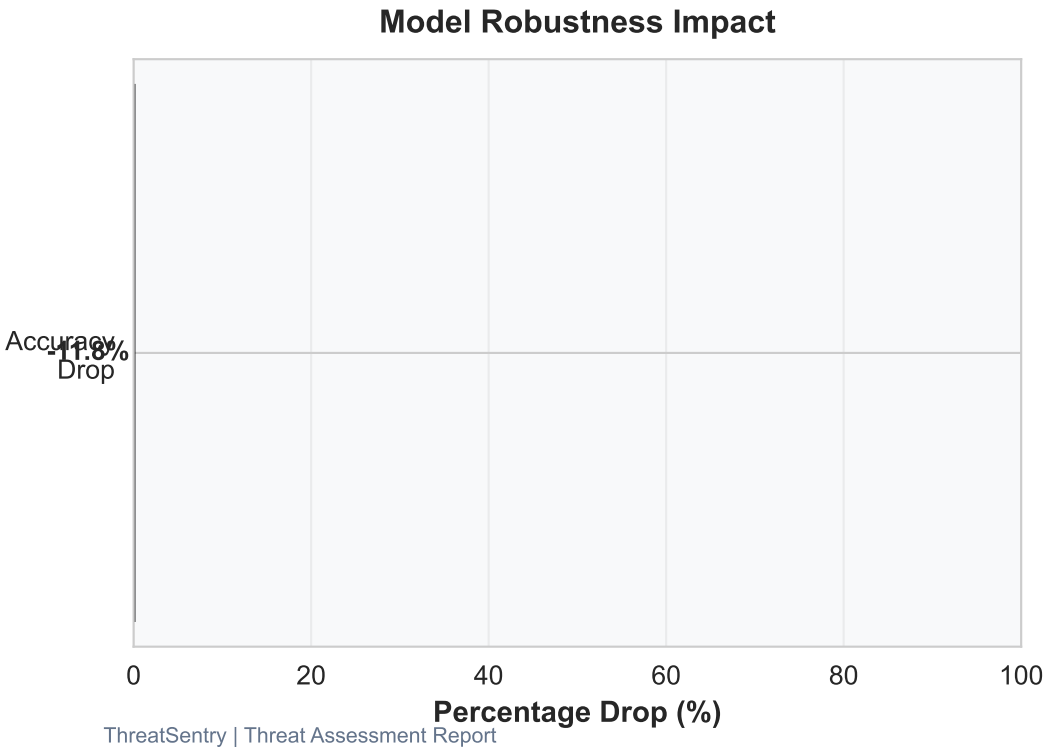
Assessment Narrative

Successfully executed FGSM attack on model 20260205_171124_small_image_classifier.pt using 10 test images. Attack success rate: 100.0%. Average original accuracy: 36.89%, Average adversarial accuracy: 48.71%. The attack successfully fooled the model in 10 out of 10 cases.

Detailed Analysis







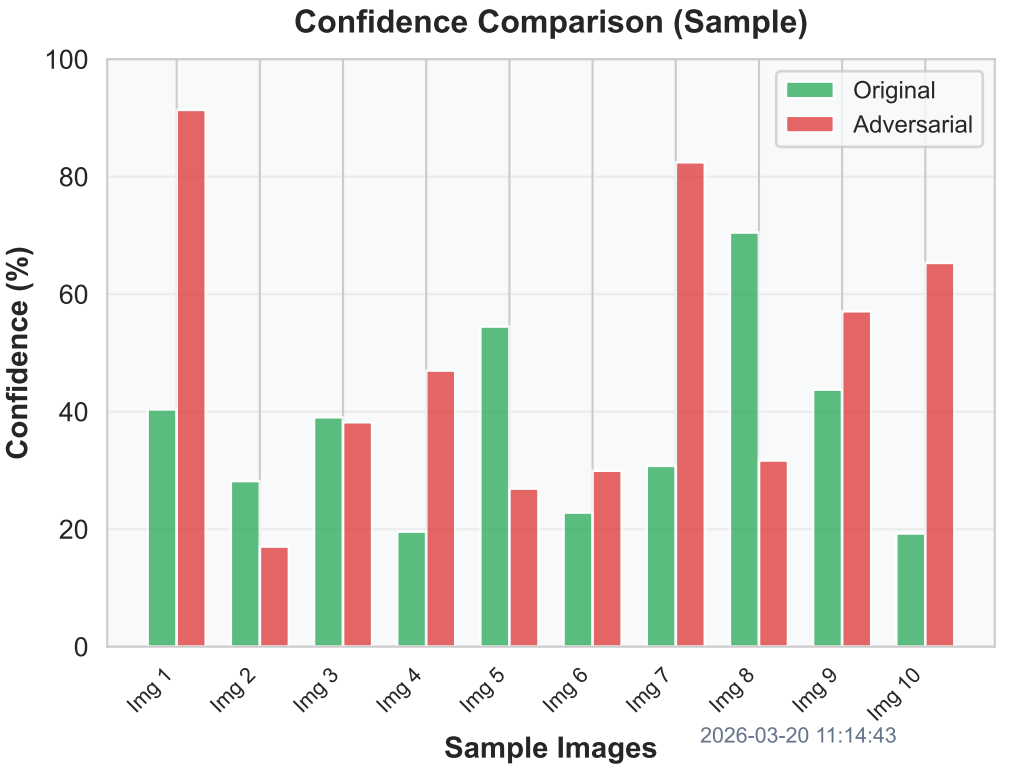
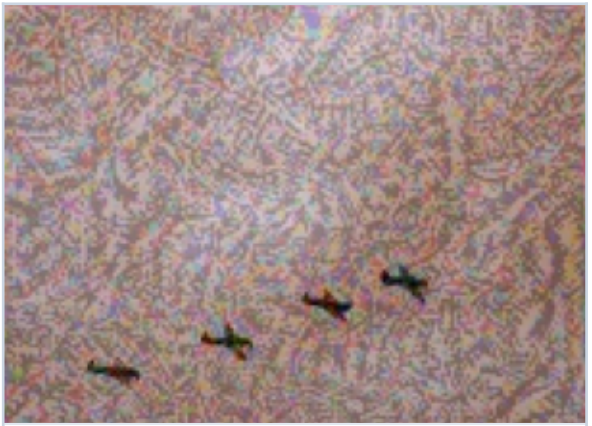
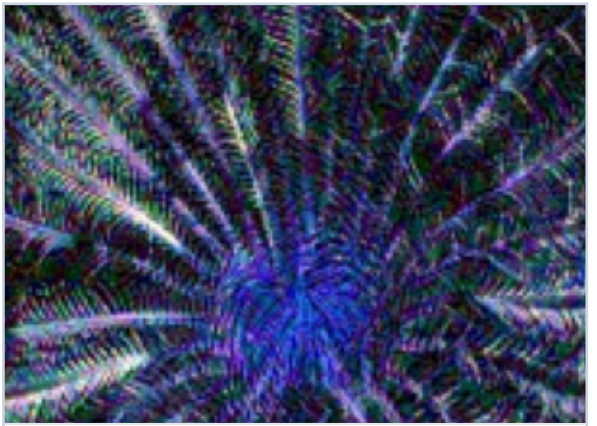


Image Evidence and Predictions

Images 1-4 of 10



Attack Successful
Original image: warplane.JPEG
Adversarial image: warplane.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 109
Original confidence: 54.51%
After attack predicted as: Class 824
Adversarial confidence: 26.91%



Attack Successful
Original image: sea_urchin.JPEG
Adversarial image: sea_urchin.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 84
Original confidence: 40.39%
After attack predicted as: Class 815
Adversarial confidence: 91.38%

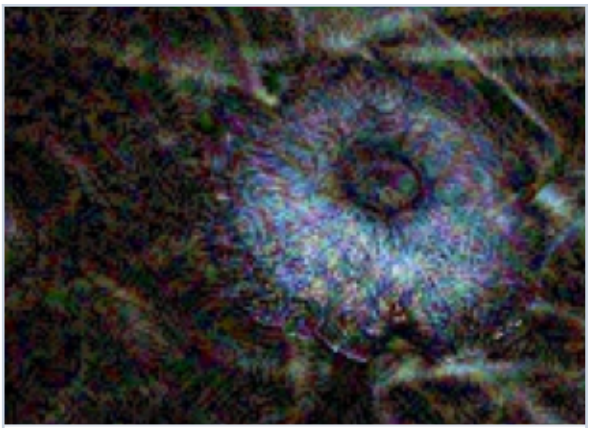
Attack Successful
Original image: warthog.JPEG
Adversarial image: warthog.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 342
Original confidence: 70.50%
After attack predicted as: Class 815
Adversarial confidence: 31.70%

Attack Successful
Original image: Irish_wolfhound.JPEG
Adversarial image: Irish_wolfhound.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 170
Original confidence: 22.84%
After attack predicted as: Class 815
Adversarial confidence: 29.96%

Image Evidence and Predictions

Attack Successful
Original image: bloodhound.JPEG
Adversarial image: bloodhound.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 282
Original confidence: 43.77%
After attack predicted as: Class 815
Adversarial confidence: 57.10%

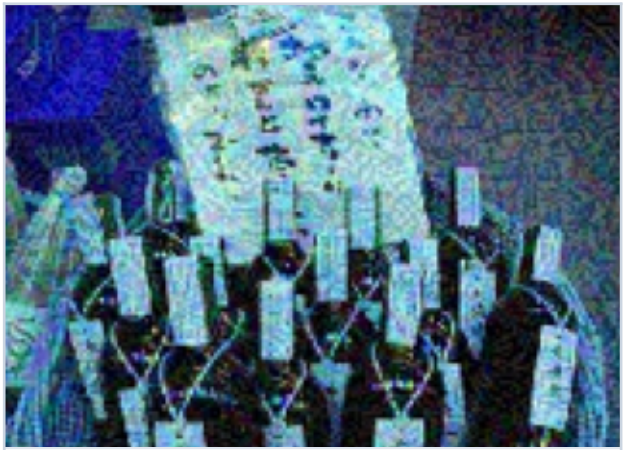
Attack Successful
Original image: white_wolf.JPEG
Adversarial image: white_wolf.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 363
Original confidence: 30.79%
After attack predicted as: Class 815
Adversarial confidence: 82.46%



Attack Successful
Original image: mushroom.JPEG
Adversarial image: mushroom.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 297
Original confidence: 19.27%
After attack predicted as: Class 815
Adversarial confidence: 65.34%

Attack Successful
Original image: basenji.JPEG
Adversarial image: basenji.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 608
Original confidence: 39.06%
After attack predicted as: Class 611
Adversarial confidence: 38.21%

Image Evidence and Predictions



Attack Successful
Original image: wine_bottle.JPEG
Adversarial image: wine_bottle.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 71
Original confidence: 28.19%
After attack predicted as: Class 815
Adversarial confidence: 17.05%

Attack Successful
Original image: tiger.JPEG
Adversarial image: tiger.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: Class 362
Original confidence: 19.60%
After attack predicted as: Class 109
Adversarial confidence: 47.01%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model 20260205_171124_small_image_classifier.pt using 10 test images. Attack success rate: 100.0%. Average original accuracy: 36.89%, Average adversarial accuracy: 48.71%. The attack successfully fooled the model in 10 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques